

Chapter 2

Ramification After Blowup Is Tame

Throughout this chapter, let C be a smooth, projective, geometrically connected curve over a field k of characteristic $p > 0$. Fix a modulus

$$\mathfrak{m} = \sum_i n_i P_i, \quad U = C \setminus \mathfrak{m}.$$

For a single-point submodule $\mathfrak{n} = n_i P_i$, put $d_{\mathfrak{n}} = \deg(\mathfrak{n})$. The divisor $\mathfrak{n}$ determines a k -rational point $Z_{\mathfrak{n}} \in C^{(d_{\mathfrak{n}})}$. We write

$$\pi_{\mathfrak{n}} : X_{\mathfrak{n}} := \mathrm{Bl}_{Z_{\mathfrak{n}}}(C^{(d_{\mathfrak{n}})}) \longrightarrow C^{(d_{\mathfrak{n}})}$$

for the blowup, $E_{\mathfrak{n}}$ for its exceptional divisor, and $\eta_{\mathfrak{n}}$ for the generic point of $E_{\mathfrak{n}}$. By [Propositions 22](#) and [31](#), $E_{\mathfrak{n}} \cong \mathbb{P}_k^{d_{\mathfrak{n}}-1}$ and in particular is an irreducible prime divisor. Thus

$$\begin{array}{ccc} E_{\mathfrak{n}} = \overline{\{\eta_{\mathfrak{n}}\}} & \hookrightarrow & X_{\mathfrak{n}} \\ \downarrow & & \downarrow \pi_{\mathfrak{n}} \\ Z_{\mathfrak{n}} & \hookrightarrow & C^{(d_{\mathfrak{n}})}. \end{array} \tag{2.1}$$

The purpose of the chapter is to prove [Theorem 3](#). If $\mathcal{F}$ is the rank-one Λ -local system in that theorem and $\mathcal{P} = \mathrm{Fr}(\mathcal{F})$ is its $G = \Lambda^{\times}$ -torsor of frames, then [Proposition 9](#) and [definition 26](#) gives

$$\mathrm{Fr}(\mathcal{F}^{[d_{\mathfrak{n}}]}) \cong \mathcal{P}^{[d_{\mathfrak{n}}]}.$$

The local system and its frame torsor therefore have the same ramification. It is enough to prove the following torsor statement.

Theorem 33. *Let G be a finite abelian group and let $\mathcal{P} \rightarrow U$ be a G -torsor whose ramification is bounded by $\mathfrak{m}$. For every single-point submodule $\mathfrak{n} = n_i P_i$, the symmetric-power torsor $\mathcal{P}^{[d_{\mathfrak{n}}]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

The organization is deliberately parallel. We first define ramification for finite covers and torsors. We then place Kummer, Artin–Schreier, and Artin–Schreier–Witt theory next to one another, separating this general background from the calculation particular to the thesis. A common prelude collects the formal-local reduction, the contracted-product mechanism, and the geometry of the exceptional valuation. Finally, the three cyclic calculations are presented in the same order and with the same internal steps. The general finite abelian case follows by primary decomposition.

2.1 Ramification of Finite Covers and Torsors

2.1.1 Discrete valuation fields

Definition 34 ([Stacks, Tag 09E4]). An extension of discrete valuation rings is an injective local homomorphism $A \subset B$ of discrete valuation rings. If π_A and π_B are uniformizers, then $\pi_A = u\pi_B^e$ for a unit $u \in B$ and a uniquely determined integer $e \geq 1$. The integer e is the *ramification index*. The extension is *weakly unramified* if $e = 1$.

Let A have fraction field K , let L/K be finite separable, and let B be the integral closure of A in L . For the maximal ideals $\mathfrak{m}_i$ of B above $\mathfrak{m}_A$, write e_i and f_i for the ramification indices and residue degrees. Then

$$[L : K] = \sum_i e_i f_i.$$

When A is henselian, there is only one prime above $\mathfrak{m}_A$.

Definition 35 ([Stacks, Tag 09E9]). With the notation above, L/K is

1. *unramified* if every $e_i = 1$ and every residue extension is separable;
2. *tamely ramified* if every residue extension is separable and, when the residue characteristic is $p > 0$, every e_i is prime to p ;
3. *totally ramified* if there is one prime above $\mathfrak{m}_A$ and its residue extension is trivial.

Now suppose A is complete with perfect residue field, L/K is finite Galois, and $G = \text{Gal}(L/K)$. Let B be the valuation ring of L and v_B its valuation. The lower ramification groups are

$$G_i = \{\sigma \in G : v_B(\sigma(b) - b) \geq i + 1 \text{ for every } b \in B\}, \quad i \geq -1.$$

Thus $G_{-1} = G$, G_0 is inertia, L/K is unramified exactly when $G_0 = 1$, and it is tame exactly when $G_1 = 1$. The Herbrand function

$$\phi_{L/K}(u) = \int_0^u \frac{dt}{[G_0 : G_t]}$$

reindexes the filtration by $G^v = G_{\phi_{L/K}^{-1}(v)}$. The reason for using upper numbering in this thesis is its compatibility with quotients: if $H \triangleleft G$, then

$$(G/H)^v = G^v H/H.$$

2.1.2 Ramification along a divisor

Let X be locally Noetherian, let $U \subset X$ be dense, and let $Y \rightarrow U$ be finite etale. If $Z \subset X \setminus U$ is a prime divisor with generic point ξ and $\mathcal{O}_{X,\xi}$ is a discrete valuation ring, put $K_\xi = \text{Frac}(\mathcal{O}_{X,\xi})$. The pullback of Y to $\text{Spec } K_\xi$ is the spectrum of a finite separable K_ξ -algebra L_ξ .

Definition 36. The cover $Y \rightarrow U$ is *unramified*, respectively *tamely ramified*, at (Z, ξ) if every field factor of L_ξ is unramified, respectively tamely ramified, with respect to $\mathcal{O}_{X,\xi}$. It is unramified, respectively tamely ramified, in codimension one if this holds at every prime divisor in $X \setminus U$.

Return to the curve C . For $P \in C \setminus U$, put $\hat{K}_P = \text{Frac}(\hat{\mathcal{O}}_{C,P})$. If $\mathcal{P} \rightarrow U$ is a G -torsor, then

$$\mathcal{P}|_{\text{Spec } \hat{K}_P} \cong \coprod_{G/H} \text{Spec } F,$$

where $F/\hat{K}_P$ is Galois and $H \subseteq G$ is the stabilizer of a connected component.

Definition 37. The extension $F/\widehat{K}_P$ has *ramification bounded by r* if $H^r = 1$. The torsor $\mathcal{P}$ has *ramification bounded by r at P* if its connected field factors have this property. It has *ramification bounded by $\mathfrak{m} = \sum n_i P_i$* if its ramification at every P_i is bounded by n_i .

Equivalently, the local torsor is classified by a continuous homomorphism

$$\rho_P : G_{\widehat{K}_P} \longrightarrow G,$$

and the bound is the condition $\rho_P(G_{\widehat{K}_P}^r) = 1$. This character formulation also shows that the definition is unchanged after passing to the geometric points over P .

2.1.3 Operations on ramified torsors

Lemma 38. Let G be finite abelian and let $\mathcal{P}_1, \mathcal{P}_2$ be G -torsors on a dense open $U \subset X$. If their ramification at a prime divisor is bounded by r_1 and r_2 , then the ramification of $\mathcal{P}_1 \otimes^G \mathcal{P}_2$ is bounded by $\max(r_1, r_2)$.

Proof. Over the completed local field the two torsors correspond to characters $\rho_1, \rho_2 : G_K \rightarrow G$, while their contracted product corresponds to $\rho_1 + \rho_2$. Both characters vanish on $G_K^{\max(r_1, r_2)}$, so their sum does as well. $\square$

Lemma 39. Let $f : X \rightarrow Y$ carry a prime divisor (Z_X, ξ_X) to a prime divisor (Z_Y, ξ_Y) and be etale at ξ_X . If $\mathcal{P}$ is a finite abelian torsor on a dense open of Y , then $f^{-1}\mathcal{P}$ has ramification bounded by r at ξ_X if and only if $\mathcal{P}$ has ramification bounded by r at ξ_Y .

Proof. The induced extension of completed discrete valuation fields is unramified. The upper-numbering filtration of the absolute Galois group is preserved under unramified base change, so the two local characters kill the same r -th ramification group. $\square$

If $H \subseteq G$, the quotient $\mathcal{P}/H$ corresponds locally to the composite $G_K \rightarrow G \rightarrow G/H$. Consequently ramification bounds descend to quotient torsors. Together with the preceding two lemmas, this is the functoriality used in the final reduction from cyclic groups to arbitrary finite abelian groups.

2.2 Three Parallel Theories of Cyclic Torsors

This section contains general background, not the symmetric-power calculation specific to the thesis. The three theories have the same formal shape:

theory	H	$\varphi : H \rightarrow H$	$\ker(\varphi)$
Kummer	$\mathbb{G}_m$	$z \mapsto z^e$	$\mu_e \cong \mathbb{Z}/e\mathbb{Z}$
Artin–Schreier	$\mathbb{G}_a$	$z \mapsto z^p - z$	$\mathbb{F}_p \cong \mathbb{Z}/p\mathbb{Z}$
Artin–Schreier–Witt	$\mathbb{W}_r$	$F - \text{id}$	$W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$

In every row, pulling back φ along a section a produces a torsor $\varphi(X) = a$, and changing a by an element of $\varphi(H)$ does not change its class.

2.2.1 Kummer theory: order prime to p

Let e be prime to p and suppose the field K contains μ_e . The Kummer sequence

$$1 \longrightarrow \mu_e \longrightarrow \mathbb{G}_m \xrightarrow{(-)^e} \mathbb{G}_m \longrightarrow 1$$

identifies

$$H^1(K, \mu_e) \cong K^\times / (K^\times)^e.$$

The class of $a \in K^\times$ is represented by $K[T]/(T^e - a)$, with $\zeta \in \mu_e$ acting by $T \mapsto \zeta T$.

Theorem 40 (Ramification in Kummer extensions, [Koc97, Proposition 1.83]). *Let K be a discretely valued field containing μ_e , with residue characteristic prime to e , and let $L = K[T]/(T^e - a)$. Write $a = \pi^m u$ with u a unit. Then the ramification index is*

$$e(L/K) = \frac{e}{\gcd(e, m)}.$$

If $e \mid m$, the extension is unramified or trivial according to the residue class of u ; if $\gcd(e, m) = 1$, it is totally and tamely ramified of degree e . Conversely, every cyclic extension of degree prime to p over a field containing the required roots of unity is Kummer.

2.2.2 Artin–Schreier theory: order p

For a field K of characteristic p , the Artin–Schreier sequence

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbb{G}_a \xrightarrow{\wp} \mathbb{G}_a \longrightarrow 0, \quad \wp(z) = z^p - z,$$

identifies

$$H^1(K, \mathbb{Z}/p\mathbb{Z}) \cong K/\wp(K).$$

The class of $a \in K$ is represented by $K[X]/(X^p - X - a)$, with $g \in \mathbb{F}_p$ acting by $X \mapsto X + g$.

Theorem 41 (Ramification in Artin–Schreier extensions, [Tho05]). *Let $K = k((x))$, with k perfect of characteristic p , and let a be a reduced representative of a class in $K/\wp(K)$: either a is regular, or $v_x(a) = -m < 0$ with $p \nmid m$. If the class is nonzero and regular, the corresponding extension is unramified. If $v_x(a) = -m < 0$, it is totally ramified of degree p and its unique upper ramification break is m . Every cyclic extension of degree p arises in this way.*

2.2.3 Artin–Schreier–Witt theory: order p^r

We recall only the Witt-vector structure used later. For $n \geq 0$, the n -th p -typical Witt polynomial is

$$w_n(X_0, \dots, X_n) = \sum_{j=0}^n p^j X_j^{p^{n-j}}.$$

There are unique integral polynomials defining addition and multiplication on A^r for which the ghost map $(w_0, \dots, w_{r-1})$ is a ring map. The resulting ring is denoted $W_r(A)$.

Theorem 42 (Witt; [Ser79, II, §6]). *The functor $A \mapsto W_r(A)$ is a commutative ring functor. Witt addition is given by universal polynomials S_n , and the n -th component has triangular form*

$$(\vec{a} \oplus \vec{b})_n = a_n + b_n + C_n(a_0, \dots, a_{n-1}; b_0, \dots, b_{n-1}). \quad (2.2)$$

The ghost map is injective when p is a non-zero-divisor and bijective when p is invertible.

Lemma 43 (Ghost principle). *Two universal polynomial operations on Witt vectors are equal if their ghost components agree over an integral polynomial ring.*

Proof. Over that polynomial ring p is a non-zero-divisor, so the ghost map is injective. The resulting polynomial identity remains true after arbitrary base change. $\square$

Proposition 44 (Shape of Witt addition). *The carry polynomial C_n in (2.2) involves only components of index strictly less than n . The analogous statement holds for Witt subtraction $\ominus$.*

Proof. Solve recursively for S_n in the identity $w_n(S_0, \dots, S_n) = w_n(\vec{X}) + w_n(\vec{Y})$. The terms containing X_n, Y_n contribute $p^n(X_n + Y_n)$; all remaining terms involve lower components. $\square$

For an $\mathbb{F}_p$ -algebra A , Frobenius is componentwise, $F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p)$. Verschiebung and truncation are

$$V(a_0, \dots, a_{r-2}) = (0, a_0, \dots, a_{r-2}), \quad t(a_0, \dots, a_{r-1}) = (a_0, \dots, a_{r-2}).$$

Definition 45. We use F , V , and t for these three operators on finite-length p -typical Witt vectors.

Proposition 46. *The map t is a ring homomorphism with kernel $V^{r-1}A$; the map V is additive and commutes with F ; and multiplication by p is $VF = FV$. In particular,*

$$p^{r-1}(a_0, \dots, a_{r-1}) = (0, \dots, 0, a_0^{p^{r-1}}). \quad (2.3)$$

Example 47. The additive group $W_r(\mathbb{F}_p)$ is canonically $\mathbb{Z}/p^r\mathbb{Z}$; the unit $(1, 0, \dots, 0)$ is a generator.

Definition 48. The Artin–Schreier–Witt operator is the additive endomorphism

$$\wp = F - \text{id} : W_r(A) \longrightarrow W_r(A).$$

For $r = 1$ this is $z \mapsto z^p - z$.

Lemma 49 (Witt bookkeeping). *For an $\mathbb{F}_p$ -algebra A :*

- (i) t commutes with F and $\wp$, and $\ker(t) = V^{r-1}A$;
- (ii) V^{r-1} is additive and $\wp V^{r-1} = V^{r-1}\wp$;
- (iii) addition in the top component is carry-free:

$$\vec{a} \oplus V^{r-1}(c) = (a_0, \dots, a_{r-2}, a_{r-1} + c).$$

Proof. The first two assertions follow from the preceding proposition. For the last, compare ghost components: $V^{r-1}(c)$ has zero ghost components below level $r - 1$ and last ghost component $p^{r-1}c$. $\square$

The Witt-vector group scheme $\mathbb{W}_r$ has underlying scheme $\mathbb{A}_{\mathbb{F}_p}^r$ and group law given by Witt addition. The operator $\wp$ gives the exact sequence

$$0 \longrightarrow \mathbb{Z}/p^r\mathbb{Z} \cong W_r(\mathbb{F}_p) \longrightarrow \mathbb{W}_r \xrightarrow{\wp} \mathbb{W}_r \longrightarrow 0. \quad (2.4)$$

Lemma 50 (ASW covers are etale torsors). *Let A be an $\mathbb{F}_p$ -algebra and $\vec{g} \in W_r(A)$. Then*

$$B = A[\vec{X}]/(\wp\vec{X} \ominus \vec{g})$$

is finite free of rank p^r and etale over A . Translation $\vec{X} \mapsto \vec{X} \oplus \underline{m}$ by $m \in W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$ makes $\text{Spec } B$ a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor.

Proof. By triangularity, the n -th equation is monic of degree p in X_n and only involves lower variables otherwise. Hence the monomials $\prod X_n^{a_n}$, $0 \leq a_n < p$, form a basis. The Jacobian is triangular with diagonal entries -1 , so the algebra is etale. Its translations are the kernel $W_r(\mathbb{F}_p)$ of $\wp$. $\square$

Theorem 51 (Artin–Schreier–Witt theory; [Tho05, §§3–4]). *For a field K of characteristic p there is a canonical isomorphism*

$$W_r(K)/\wp W_r(K) \xrightarrow{\sim} H^1(K, \mathbb{Z}/p^r\mathbb{Z}) = \text{Hom}_{\text{cont}}(G_K, \mathbb{Z}/p^r\mathbb{Z}).$$

The class of $\vec{g}$ corresponds to the torsor $\wp\vec{X} = \vec{g}$. Its order is the order of the image of the associated character; in particular, the torsor is connected of degree p^r exactly when $[\vec{g}]$ has order p^r . Every cyclic extension of degree dividing p^r arises this way.

Corollary 52. *If $\vec{g} = (g_0, \dots, g_{r-1})$ has $g_0 = 0$, then $p^{r-1}[\vec{g}] = 0$. Consequently every representative of a class of order p^r has nonzero first component.*

Proof. Apply (2.3). $\square$

Remark. If $K = k((x))$ and k is algebraically closed, every finite extension of K is totally ramified. Hence a class of order p^r defines a cyclic totally ramified extension of degree p^r .

Lemma 53 (Isobaricity of Witt addition). *Give the j -th component weight p^j . Every monomial in the n -th component of a Witt sum has total weight p^n . The same is true for an iterated Witt sum.*

Proof. Each ghost polynomial w_n is isobaric of weight p^n . The recursion defining S_n preserves that weight, and the assertion for iterated sums follows by substitution. $\square$

Definition 54. For $K = k((x))$, put $m(g) = \max(0, -v_x(g))$. A vector $\vec{g} = (g_0, \dots, g_{r-1}) \in W_r(K)$ is *reduced* if every nonzero pole order $m(g_j)$ is prime to p .

Theorem 55 (Schmid–Witt break formula). *Let $K = k((x))$, with k perfect, and let a reduced vector $\vec{g}$ represent a cyclic totally ramified extension of degree p^r . Its largest upper ramification break is*

$$u = \max_{0 \leq j < r} p^{r-1-j} m(g_j).$$

For $r = 1$ this is the Artin–Schreier break formula. The weighted maximum is the exact feature that replaces the single pole-order bound in the degree- p case.

2.3 Prelude to the Explicit Ramification Calculations

This section isolates everything shared by the three calculations. Its main point is that Kummer multiplication, Artin–Schreier addition, and Witt addition are instances of a single contracted-product construction.

2.3.1 A common torsor equation

Let S be a scheme, let H be a commutative S -group scheme, and let $\varphi : H \rightarrow H$ be a finite etale surjective homomorphism with constant kernel G . For a section $a : X \rightarrow H$ over an S -scheme X , write

$$\mathcal{T}_a := X \times_{a, H, \varphi} H.$$

It is a G -torsor on X . We write $\star$ for the group law of H ; thus $\star$ means multiplication for $H = \mathbb{G}_m$, ordinary addition for $H = \mathbb{G}_a$, and Witt addition for $H = \mathbb{W}_r$.

Proposition 56 (The common cyclic product calculation). *Let $a_i : X \rightarrow H$, $1 \leq i \leq d$, be sections. There is a canonical isomorphism of G -torsors*

$$\mathcal{T}_{a_1} \otimes^G \cdots \otimes^G \mathcal{T}_{a_d} \cong \mathcal{T}_{a_1 \star \cdots \star a_d}.$$

On equation coordinates it is induced by

$$(X_1, \dots, X_d) \mapsto Y := X_1 \star \cdots \star X_d, \quad \varphi(Y) = a_1 \star \cdots \star a_d.$$

If a group Γ permutes the indices, then Y and the resulting datum are Γ -invariant, so the isomorphism is compatible with descent to the quotient by Γ .

Proof. The product $\prod_i \mathcal{T}_{a_i}$ is a G^d -torsor. Contracting along the sum map $G^d \rightarrow G$ gives its quotient by the kernel K_G . The displayed map is K_G -invariant because the kernel translations cancel under $\star$; it is equivariant for the residual G -action. Its target is a G -torsor because it is the pullback of φ . Hence it is an isomorphism by [Proposition 5](#). Commutativity and associativity of $\star$ give the permutation-equivariance. $\square$

The three concrete instances are therefore

G	Y	new defining datum
$\mathbb{Z}/e\mathbb{Z}$	$\prod_i X_i$	$\prod_i a(x_i)$
$\mathbb{Z}/p\mathbb{Z}$	$\sum_i X_i$	$\sum_i a(x_i)$
$\mathbb{Z}/p^r\mathbb{Z}$	$\bigoplus_i \vec{X}_i$	$\bigoplus_i \vec{a}(x_i)$.

This is the precise sense in which the three symmetric-power calculations are the same calculation.

2.3.2 Formal-local invariance and the exceptional valuation

From now on assume k is algebraically closed. This is the setting in which the theorem is used later, and it makes every point of C rational. Fix $P \in C(k)$, put $\mathfrak{n} = dP$, and choose a uniformizer x at P . Let $x_1, \dots, x_d$ be the induced parameters on C^d , and let $e_1, \dots, e_d$ be their elementary symmetric polynomials. Set

$$V = \operatorname{Spec} k[[e_1, \dots, e_d]][e_d^{-1}], \quad W = \operatorname{Spec} k[[x_1, \dots, x_d]][(x_1 \cdots x_d)^{-1}].$$

For each i , let $q_i : W \rightarrow \operatorname{Spec} k((x))$ be given by $x \mapsto x_i$.

Lemma 57 (Formal-local invariance). *With the notation above:*

1. $\widehat{\mathcal{O}}_{C^{(d)}, dP} \cong k[[e_1, \dots, e_d]]$, and the preimage of $U^{(d)}$ is V ;
2. if η_{dP} is the generic point of the exceptional divisor, then

$$\widehat{\mathcal{O}}_{X_{dP}, \eta_{dP}} \cong \kappa[[e_d]], \quad \widehat{K}_{\eta_{dP}} \cong \kappa((e_d)), \quad \kappa = k(e_1/e_d, \dots, e_{d-1}/e_d); \quad (2.5)$$

3. for every finite abelian G -torsor $\mathcal{P} \rightarrow U$, with local restriction $\mathcal{P}_x$ to $\operatorname{Spec} k((x))$, there is a canonical isomorphism over V

$$\mathcal{P}^{[d]}|_V \cong \Xi \backslash (q_1^{-1} \mathcal{P}_x \times_W \cdots \times_W q_d^{-1} \mathcal{P}_x).$$

Consequently the ramification of $\mathcal{P}^{[d]}$ at η_{dP} depends only on d and the isomorphism class of $\mathcal{P}_x$.

Proof. Completion of C^d at $(P, \dots, P)$ is $k[[x_1, \dots, x_d]]$. Taking S_d -invariants commutes with the flat completion and gives

$$k[[x_1, \dots, x_d]]^{S_d} = k[[e_1, \dots, e_d]].$$

The boundary divisor consisting of cycles containing P pulls back to $V(x_1 \cdots x_d) = V(e_d)$, proving (1).

Blowups commute with flat base change. On the blowup chart where the e_d -direction is nonzero, write $e_i = (e_i/e_d)e_d$ for $i < d$. At the generic point of the exceptional divisor the ratios e_i/e_d are algebraically independent residue parameters and e_d is a uniformizer. This gives (2).

Finally, the formation of the finite quotient $\mathcal{P}^{[d]} = \Xi \backslash \mathcal{P}^d$ commutes with flat base change: the quotient ring is an invariant subring, hence a kernel, and kernels commute with flat base change. Pulling $\mathcal{P}^d$ to W gives the displayed product of the local torsors. Restricting further along $\text{Spec } \kappa((e_d)) \rightarrow V$ proves the consequence. $\square$

Corollary 58 (Reduction to $\mathbb{P}^1$). *Suppose $\mathcal{P} \rightarrow C \setminus \mathfrak{m}$ and $\mathcal{P}' \rightarrow C' \setminus \mathfrak{m}'$ are G -torsors and their restrictions to the punctured formal discs at rational points P, P' are isomorphic after matching uniformizers. Then $\mathcal{P}^{[d]}$ and $\mathcal{P}'^{[d]}$ have identical ramification at η_{dP} and $\eta_{dP'}$.*

Proof. Both local symmetric-power torsors are obtained from the same local torsor by the construction in Lemma 57. $\square$

We will repeatedly use the following valuation interpretation. If a symmetric Laurent polynomial belongs to

$$\kappa = k(e_1/e_d, \dots, e_{d-1}/e_d),$$

then it is a unit or zero in $\kappa[[e_d]]$ and hence has no pole along the exceptional divisor. By contrast, a power e_d^a has valuation a .

2.3.3 Polynomial normal forms on $\mathbb{G}_m$

The formal-local reduction becomes useful once a local cyclic torsor is realized by an equation on $\mathbb{G}_m$.

Lemma 59 (Surjectivity on integral Witt vectors). *For algebraically closed k , the map $\wp : W_r(k[[x]]) \rightarrow W_r(k[[x]])$ is surjective.*

Proof. For $r = 1$, solve $u^p - u = h$ coefficientwise; algebraic closedness solves the constant term. Induct on r . Solve after truncation, lift the solution to length r , and use Lemma 49 to correct the remaining top component by an element of $V^{r-1}k[[x]]$. $\square$

Definition 60. A Witt vector $\vec{f} = (f_0, \dots, f_{r-1}) \in W_r(k((x)))$ is a *reduced polynomial vector* if it is reduced and every $f_j \in x^{-1}k[x^{-1}]$. Put $m_j = \deg_{x^{-1}} f_j$, with $m_j = 0$ when $f_j = 0$.

Proposition 61 (Reduced polynomial representatives). *Every class in $W_r(k((x)))/\wp W_r(k((x)))$ has a reduced polynomial representative.*

Proof. Induct on r . For $r = 1$, remove the integral part using surjectivity on $k[[x]]$. If the remaining leading pole cx^{-m} has $p \mid m$, subtract

$$\wp(c^{1/p}x^{-m/p}) = cx^{-m} - c^{1/p}x^{-m/p},$$

this strictly lowers the pole order, and iteration produces a reduced polynomial.

For the induction step, first reduce the truncation to length $r - 1$ and lift the change of representative. The lower components are now reduced polynomials. By [Lemma 49](#), subtracting $\wp(V^{r-1}z)$ changes only the last component, by $-\wp(z)$. Apply the length-one procedure to that component. Every operation subtracts an actual element of $\wp W_r(k((x)))$, so the class is unchanged. $\square$

Lemma 62 (Parallel realization over $\mathbb{G}_m$). *Let k be algebraically closed and let $\mathcal{Q}$ be a connected cyclic torsor on $\text{Spec } k((x))$.*

1. *If $G = \mathbb{Z}/e\mathbb{Z}$, $(e, p) = 1$, then for some a prime to e the torsor is represented by*

$$T^e = x^a.$$

The same equation defines a torsor on $\mathbb{G}_m$.

2. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification is bounded by d , then the torsor is represented by*

$$X^p - X = f(x), \quad f \in x^{-1}k[x^{-1}], \quad m := \deg_{x^{-1}} f < d, \quad p \nmid m,$$

with the convention $f = 0$ for the trivial class. The same equation defines a torsor on $\mathbb{G}_m$ extending etale across ∞ .

3. *If $G = \mathbb{Z}/p^r\mathbb{Z}$ and the ramification is bounded by d , then the torsor is represented by*

$$\wp \vec{X} = \vec{f},$$

where $\vec{f}$ is a reduced polynomial vector satisfying

$$p^{r-1-j}m_j < d \quad (0 \leq j < r). \quad (2.6)$$

The same equation defines a torsor on $\mathbb{G}_m$ extending etale across ∞ .

More generally, a tamely ramified G -torsor for finite abelian G is induced from the Kummer torsor in (1) along an injection $\mathbb{Z}/e\mathbb{Z} \hookrightarrow G$.

Proof. For (1), $k((x))$ has no nontrivial unramified extension and its tame quotient is procyclic. Kummer theory identifies its connected quotient of order e with the class of x^a for $a \in (\mathbb{Z}/e\mathbb{Z})^\times$.

For (2), Artin–Schreier theory identifies the class with an element of $k((x))/\wp(k((x)))$. The length-one reduction in the proof of [Proposition 61](#) produces f of the stated form. The break formula gives $m < d$. Since f is a polynomial in x^{-1} without constant term, the equation is etale over $k[x^{-1}]$.

For (3), choose a reduced polynomial representative using [Proposition 61](#). Connectedness and [Corollary 52](#) imply $f_0 \neq 0$. The Schmid–Witt formula gives

$$\max_j p^{r-1-j}m_j < d,$$

which is (2.6). The torsor and its extension across ∞ follow from [Lemma 50](#) applied over $k[x, x^{-1}]$ and $k[x^{-1}]$. $\square$

Remark. The Witt pole bound forces $f_j = 0$ whenever $p^{r-1-j} \geq d$. Thus the strong restriction imposed by “break $< d$ ” is visible in every Witt component, not only in the last one.

2.3.4 Symmetric regularity

Put $y_i = x_i^{-1}$. The identity

$$e_j(y_1, \dots, y_d) = \frac{e_{d-j}(x_1, \dots, x_d)}{e_d(x_1, \dots, x_d)}$$

relates symmetric polynomials in the poles y_i to the residue field κ of the exceptional valuation.

Lemma 63 (Symmetric regularity). *If $\Phi \in k[y_1, \dots, y_d]^{S_d}$ and every monomial of Φ has total degree strictly less than d , then*

$$\Phi \in k(e_1/e_d, \dots, e_{d-1}/e_d) = \kappa.$$

In particular, Φ is regular at η_{dP} .

Proof. Expand Φ in monomial symmetric functions. A symmetric function of degree $< d$ is a polynomial in $e_1(y), \dots, e_{d-1}(y)$; $e_d(y)$ cannot occur. The displayed identity puts each of these generators in κ . $\square$

2.3.5 The scheme-theoretic symmetric power causes no new divisor

The torsor-theoretic symmetric power is a quotient of the ordinary symmetric power of the covering scheme. The comparison is harmless in codimension one.

Lemma 64. *Let $\mathcal{P} = U' \rightarrow U$ be a finite abelian G -torsor on a smooth curve, and let C' be the smooth proper model of U' . The natural morphism from the ordinary symmetric power $U'^{(d)}$ to the torsor-theoretic symmetric power $U'^{(d)}$ is unramified along every codimension-one boundary component of $C'^{(d)}$.*

Proof. A boundary component has generic point represented in $C'^{(d)}$ by $(Q, Q_2, \dots, Q_d)$, where the last $d-1$ entries are generic and independent. The quotient group is $\Xi = K_G \rtimes S_d$. An element of the inertia group at this generic point must fix each of the independent function-field factors, so its permutation part and its last $d-1$ group components are trivial. The condition that it lie in K_G then forces its first component to be trivial. Thus the inertia group is trivial, and the induced extension of discrete valuation rings is weakly unramified. $\square$

2.4 Three Parallel Explicit Calculations

We now perform the thesis-specific calculation. In all three cases we make the same reductions and carry out the same five steps:

	$\mathbb{Z}/e\mathbb{Z}$	$(\mathbb{Z}/p\mathbb{Z})$	$\mathbb{Z}/p^r\mathbb{Z}$
normal form	$T^e = x^a$	$X^p - X = f$	$\wp \vec{X} = \vec{f}$
product coordinate	$\prod_i T_i$	$\sum_i X_i$	$\bigoplus_i \vec{X}_i$
new datum	e_d^a	$\sum_i f(x_i)$	$\bigoplus_i \vec{f}(x_i)$
exceptional behavior	$v(e_d^a) = a$	regular	componentwise regular
conclusion	tame	unramified	unramified.

By [Corollary 58](#) and [lemma 62](#), it is enough in each case to take

$$C = \mathbb{P}_k^1, \quad U = \mathbb{G}_m, \quad \mathbf{n} = d \cdot 0,$$

and one of the displayed equations. The complete local field at the generic point of the exceptional divisor is always

$$\widehat{K}_\eta = \kappa((e_d)), \quad \widehat{\mathcal{O}}_\eta = \kappa[[e_d]], \quad \kappa = k(e_1/e_d, \dots, e_{d-1}/e_d).$$

Theorem 65. *In the preceding model:*

1. if $\mathcal{P}$ is a $\mathbb{Z}/e\mathbb{Z}$ -torsor with $(e, p) = 1$, then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{d,0}$;
2. if $\mathcal{P}$ is a $\mathbb{Z}/p\mathbb{Z}$ -torsor whose ramification at 0 is bounded by d , then $\mathcal{P}^{[d]}$ is unramified at $\eta_{d,0}$;
3. if $\mathcal{P}$ is a $\mathbb{Z}/p^r\mathbb{Z}$ -torsor whose ramification at 0 is bounded by d , then $\mathcal{P}^{[d]}$ is unramified at $\eta_{d,0}$.

2.4.1 The Kummer calculation: $G = \mathbb{Z}/e\mathbb{Z}$, $(e, p) = 1$

Local normal form and realization

By the realization lemma, the connected local torsor is represented by

$$T^e = x^a, \quad \gcd(a, e) = 1.$$

The same equation over $k[x, x^{-1}]$ gives a $\mathbb{Z}/e\mathbb{Z}$ -torsor on $\mathbb{G}_m$.

Contracted product and descent

Pull the equation back along the d projections from $\mathbb{G}_m^d$. By [Proposition 56](#), the contracted product has coordinate

$$Y = \prod_{i=1}^d T_i$$

and equation

$$Y^e = \prod_{i=1}^d x_i^a = (x_1 \cdots x_d)^a = e_d^a.$$

Both Y and e_d^a are S_d -invariant. Hence this equation descends from $\mathbb{G}_m^d$ to the torsor $\mathcal{P}^{[d]}$ on

$$\mathbb{G}_m^{(d)} = \operatorname{Spec} k[e_1, \dots, e_d, e_d^{-1}].$$

Restriction to the exceptional divisor

Over $\widehat{K}_\eta = \kappa((e_d))$, the equation remains

$$Y^e = e_d^a.$$

Its defining element has valuation a , and the ramification index divides e . Since $(e, p) = 1$, [Theorem 40](#) shows that every field factor is tamely ramified. Thus $\mathcal{P}^{[d]}$ is tame at η .

2.4.2 The Artin–Schreier calculation: $G = \mathbb{Z}/p\mathbb{Z}$

Local normal form and realization

The local torsor has an equation

$$X^p - X = f(x), \quad f \in x^{-1}k[x^{-1}], \quad m := \deg_{x^{-1}} f < d.$$

It extends to a torsor on $\mathbb{G}_m$ that is unramified at ∞ .

Contracted product and descent

By the common product calculation, the d -fold contracted product on $\mathbb{G}_m^d$ has coordinate

$$Y = \sum_{i=1}^d X_i$$

and equation

$$Y^p - Y = \alpha, \quad \alpha := \sum_{i=1}^d f(x_i).$$

The coordinate Y and the datum α are S_d -invariant. Therefore

$$\mathcal{P}^{[d]} \cong \operatorname{Spec} \frac{k[e_1, \dots, e_d, e_d^{-1}][Y]}{(Y^p - Y - \alpha)}.$$

Degree estimate and exceptional restriction

Writing $y_i = x_i^{-1}$, the expression α is a symmetric polynomial in $y_1, \dots, y_d$ of total degree at most $m < d$. By [Lemma 63](#),

$$\alpha \in \kappa \subset \widehat{\mathcal{O}}_\eta.$$

Consequently the equation $Y^p - Y = \alpha$ defines a finite etale algebra over $\widehat{\mathcal{O}}_\eta$: its derivative with respect to Y is -1 . Its generic fiber is the restriction of $\mathcal{P}^{[d]}$ to $\widehat{K}_\eta$, so every field factor is unramified. This proves the degree- p case.

2.4.3 The Artin–Schreier–Witt calculation: $G = \mathbb{Z}/p^r\mathbb{Z}$

Local normal form and realization

First quotient away the unramified part of the local character. The remaining inertia group is cyclic of order p^s for some $s \leq r$; since the argument treats all lengths at once, rename s as r . We may therefore assume that the local torsor is connected and totally ramified. By [Lemma 62](#), it has an equation

$$\wp \vec{X} = \vec{f} = (f_0, \dots, f_{r-1}), \quad f_j \in x^{-1}k[x^{-1}],$$

where, with $m_j = \deg_{x^{-1}} f_j$,

$$p^{r-1-j}m_j < d.$$

Contracted product and descent

Put $R = k[x, x^{-1}]$ and

$$S = R[\vec{X}]/(\wp \vec{X} \ominus \vec{f}).$$

After pulling to $\mathbb{G}_m^d$, define

$$\vec{Y} = \bigoplus_{i=1}^d \vec{X}_i, \quad \vec{\alpha} = \bigoplus_{i=1}^d \vec{f}(x_i).$$

Proposition 66 (Identification of the Witt product torsor). *Every component of $\vec{Y}$ is K_G -invariant, every component of $\vec{\alpha}$ is a symmetric Laurent polynomial, and*

$$(S^{\otimes_k d})^{K_G} \cong R^{\otimes_k d}[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

Taking S_d -invariants gives

$$((S^{\otimes kd})^{K_G})^{S_d} \cong k[e_1, \dots, e_d, e_d^{-1}][\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

Thus

$$\mathcal{P}^{[d]}|_{\mathrm{Spec} \hat{K}_\eta} \cong \mathrm{Spec} \hat{K}_\eta[\vec{Y}]/(\wp \vec{Y} \ominus \vec{\alpha}).$$

Proof. The K_G -translations telescope in the Witt sum $\vec{Y}$, so it is invariant, and additivity of $\wp$ gives $\wp \vec{Y} = \vec{\alpha}$. The resulting morphism from the displayed ASW torsor to the contracted product is equivariant between G -torsors, hence an isomorphism by [Proposition 5](#); this is also [Proposition 56](#) for $H = \mathbb{W}_r$.

Permuting the tensor factors preserves both iterated Witt sums because Witt addition is commutative and associative. Since the action on Witt vectors is componentwise, every component of $\vec{Y}$ and $\vec{\alpha}$ is invariant. The ASW algebra is free over the base with basis $\prod_n Y_n^{a_n}$, $0 \leq a_n < p$, and each basis element is invariant. Therefore taking S_d -invariants acts only on the coefficients and produces the second displayed algebra. $\square$

Weighted degree estimate

The Witt components are not ordinary componentwise sums, so the Artin–Schreier degree estimate must be weighted.

Theorem 67 (Regularity of all Witt components). *Every component α_n of $\vec{\alpha} = \bigoplus_i \vec{f}(x_i)$ belongs to κ . Consequently $\vec{\alpha} \in W_r(\hat{\mathcal{O}}_\eta)$.*

Proof. Give $f_j(x_i)$ weight p^j . By [Lemma 53](#), every monomial

$$\prod_{i,j} f_j(x_i)^{a_{ij}}$$

appearing in α_n satisfies

$$\sum_{i,j} a_{ij} p^j = p^n.$$

Writing $A_j = \sum_i a_{ij}$ and expanding in $y_i = x_i^{-1}$, its total y -degree is at most

$$\sum_j A_j m_j < d \sum_j A_j p^{j-(r-1)} = d p^{n-(r-1)} \leq d.$$

The inequality is strict. The whole component α_n is symmetric by [Proposition 66](#), so [Lemma 63](#) gives $\alpha_n \in \kappa$. $\square$

Restriction to the exceptional divisor

By [Theorem 67](#), the equation

$$\wp \vec{Y} = \vec{\alpha}$$

is defined over $\hat{\mathcal{O}}_\eta$. By [Lemma 50](#), it is a finite etale $\hat{\mathcal{O}}_\eta$ -algebra. Its generic fiber is precisely the restriction of $\mathcal{P}^{[d]}$ to $\hat{K}_\eta$, hence every field factor is unramified.

Proof of Theorem 65. The Kummer, Artin–Schreier, and Artin–Schreier–Witt subsections prove parts (1), (2), and (3), respectively. $\square$

Corollary 68. *Let $\mathcal{P} \rightarrow C \setminus \mathfrak{m}$ be a finite abelian torsor and $\mathfrak{n} = dP$ a single-point submodule.*

1. *If $G = \mathbb{Z}/p\mathbb{Z}$ and the ramification at P is bounded by d , then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*
2. *If $\mathcal{P}$ is tamely ramified at P , then $\mathcal{P}^{[d]}$ is tamely ramified at $\eta_{\mathfrak{n}}$.*

Proof. Use [Lemma 62](#) to match the punctured-formal-disc restriction with the appropriate explicit torsor on $\mathbb{G}_m$, then use [Corollary 58](#) and the first two calculations above. For a tame G -torsor, induce the cyclic Kummer calculation along the injection of its cyclic inertia group into G . $\square$

Theorem 69. *Let $G = \mathbb{Z}/p^r\mathbb{Z}$ and suppose the ramification of $\mathcal{P}$ at P is bounded by $d = \deg \mathfrak{n}$. Then $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.*

Proof. Quotient away the unramified part, apply the length- r realization in [Lemma 62](#), transfer the calculation by [Corollary 58](#), and use the Artin–Schreier–Witt calculation. $\square$

2.5 Assembly for a Finite Abelian Group

We can now prove the torsor formulation stated at the beginning of the chapter.

Proof of [Theorem 33](#). Fix $\mathfrak{n} = dP$. Choose a decomposition

$$G \cong G_{p'} \times G_p, \quad G_{p'} \cong \prod_i \mathbb{Z}/e_i\mathbb{Z} \quad ((e_i, p) = 1), \quad G_p \cong \prod_j \mathbb{Z}/p^{r_j}\mathbb{Z}.$$

Projecting the character of $\mathcal{P}$ to these cyclic factors gives cyclic torsors $\mathcal{P}_i$ and $\mathcal{Q}_j$. Conversely, $\mathcal{P}$ is their product after extending structure groups along the inclusions of the factors. Symmetric powers commute with these products because both are obtained by contracting the product torsor along the relevant homomorphism of finite abelian groups.

Every $\mathcal{P}_i$ is tamely ramified at P , so its d -th symmetric power is tame at $\eta_{\mathfrak{n}}$ by the Kummer calculation. The ramification bound on $\mathcal{P}$ descends to every quotient character; hence each $\mathcal{Q}_j$ has ramification bounded by d , and [Theorem 69](#) shows that $\mathcal{Q}_j^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$. Finally, [Lemma 38](#) shows that the product of the tame prime-to- p factors and the unramified p -primary factors is tame. This product is $\mathcal{P}^{[d]}$. $\square$

2.6 Ramification on Products of Blowups

We finish with the compatibility needed in [Chapter 3](#). Let X and Y be smooth k -schemes, let $x \in X$ and $y \in Y$ be closed points, and write

$$\tilde{X} = \mathrm{Bl}_x(X), \quad \tilde{Y} = \mathrm{Bl}_y(Y), \quad \widetilde{X \times Y} = \mathrm{Bl}_{(x,y)}(X \times_k Y).$$

Denote the generic points of the three exceptional divisors by η_X, η_Y , and $\eta_{X \times Y}$.

Proposition 70. *Let G be a finite abelian group. Let $\mathcal{G}_X$ and $\mathcal{G}_Y$ be G -torsors on open subsets of $\tilde{X}$ and $\tilde{Y}$ disjoint from the exceptional divisors. If their ramification at η_X and η_Y is bounded by r , then the external contracted product*

$$\mathrm{pr}_1^{-1} \mathcal{G}_X \otimes^G \mathrm{pr}_2^{-1} \mathcal{G}_Y$$

has ramification bounded by r at $\eta_{X \times Y}$.

Proof. The assertion is etale-local at x and y , so by smoothness it is enough to take $X = \mathbb{A}_k^n$, $Y = \mathbb{A}_k^m$, and $x = y = 0$. Write the affine coordinates as $x_1, \dots, x_n, y_1, \dots, y_m$. At the generic point of the exceptional divisor of $\mathrm{Bl}_0(\mathbb{A}_k^{n+m})$, every coordinate is a uniformizer times a unit. On the chart in which the x_1 -direction is nonzero, the completed local ring is

$$k\left(\frac{x_2}{x_1}, \dots, \frac{x_n}{x_1}, \frac{y_1}{x_1}, \dots, \frac{y_m}{x_1}\right)[[x_1]].$$

There is a dense open neighborhood of $\eta_{X \times Y}$ mapping to both $\tilde{X}$ and $\tilde{Y}$: it is the locus where neither the X -directions nor the Y -directions vanish. The induced homomorphism from the exceptional DVR of $\tilde{X}$ to the displayed DVR sends the uniformizer x_1 to a uniformizer and enlarges the residue field by purely transcendental elements. It is therefore weakly unramified with separable residue extension. The corresponding statement holds for $\tilde{Y}$ (using any y_j as uniformizer on the overlap).

Upper ramification groups are preserved by such a weakly unramified, residually separable extension. Hence both pulled-back torsors have ramification bounded by r at $\eta_{X \times Y}$. Their contracted product has the same bound by [Lemma 38](#). $\square$